

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 9- BJT INV Noise Margins

Issued: 3/31/2026

Due: 4/2/2026

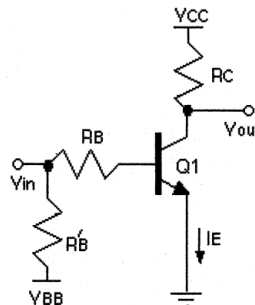
Question 1: Based on Problems 6.1 - 6.4 (D.A. Hodges and H.G. Jackson, *Analysis and Design of Digital Integrated Circuits*, 2nd Ed., McGraw-Hill, 1988.)

Objectives:

- BJT INV Noise Margin calculations

Given the NPN BJT below where:

- $\beta_F = 150$
- $V_{CC} = 5V$
- $V_{BE(on)} = 0.65V$
- $V_{BE(sat)} = 0.75V$
- $V_{CE(sat)} = 0.15V$
- $R_B = 5k\Omega$
- $R'_B = 18k\Omega$
- $R_C = 11k\Omega$
- $V_{BB} = -1.5V$



- a) Find V_{OH} , V_{OL} , V_{IH} , V_{IL} .
 V_{IH} and V_{IL} points, which are transition break points from off $\rightarrow$ FA, and FA $\rightarrow$ sat.
- b) Sketch the VTC, labelling V_{OH} , V_{OL} , V_{IH} , V_{IL} .
 - i. Plot by hand.
 - ii. Plot with LTSpice. Compare with hand results.
- c) Find I_{OH} , I_{IH} , and the fan out N, when $NM'_H = 1.75V$. HINT: $NM'_H = V'_{OH} - V_{IH}$.
 - i. Calculate by hand.
 - ii. Verify with LTSpice. Compare with hand results.

Report Submission:

Submit your work on LMS under **Week 11/Lab 9**.

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm

